

Supplementary Table 1: Complete parameter list of the NED model

| Parameter                                 | Module        | Species / Process                          | Value in notebook | Typical range                    | Status             | Basis            | Section / Code                     |
|-------------------------------------------|---------------|--------------------------------------------|-------------------|----------------------------------|--------------------|------------------|------------------------------------|
| Model control                             |               |                                            |                   |                                  |                    |                  |                                    |
| h1_auto                                   | model_control | Pathway selection switch                   | 0                 | 0 or 1                           | fixed per run      | structural       | §3                                 |
| PS scanned controls (declared scan knobs) |               |                                            |                   |                                  |                    |                  |                                    |
| ps_bd4                                    | scan_controls | Basal <i>DLL4</i> production scaler        | 0.0               | 50-150 (search scripts)          | scanned (workflow) | phenomenological | NED_model_string EC_Connect_search |
| ps_bn1                                    | scan_controls | Basal <i>NOTCH1</i> production scaler      | 0.0               | 50-150 (search scripts)          | scanned (workflow) | phenomenological | NED_model_string EC_Connect_search |
| ps_bj1                                    | scan_controls | Basal <i>Jagged1</i> production scaler     | 0.0               | 0-150 (search scripts)           | scanned (workflow) | phenomenological | NED_model_string EC_Connect_search |
| ps_j1ind (ps_ind alias)                   | scan_controls | Inducible <i>Jagged1</i> amplitude control | 0.0               | context dependent                | scanned            | phenomenological | NED_model_string EC_Connect_core   |
| ps_ja                                     | scan_controls | <i>Jagged1</i> induction threshold control | 0.0               | typically 50-5000                | scanned            | phenomenological | NED_model_string EC_Connect_search |
| ps_ni                                     | scan_controls | <i>NICD</i> activation threshold control   | 2000              | typically 2000-200000            | scanned            | phenomenological | NED_model_string EC_Connect_search |
| ps_Kdni                                   | scan_controls | <i>NICD</i> degradation control            | 0.0               | typically 0-100 (search scripts) | scanned            | phenomenological | NED_model_string EC_Connect_search |
| ps_kconv                                  | scan_controls | Productive cis conversion control          | 0.0               | typically 0-15 (search scripts)  | scanned            | phenomenological | NED_model_string EC_Connect_search |
| <i>NICD</i> module                        |               |                                            |                   |                                  |                    |                  |                                    |
| Vmni                                      | NICD_reg      | <i>NICD</i> max production                 | 25                | —                                | fixed              | phenomenological | §1.1                               |

| Parameter                        | Module                | Species / Process                              | Value in notebook            | Typical range           | Status  | Basis                                   | Section / Code      |
|----------------------------------|-----------------------|------------------------------------------------|------------------------------|-------------------------|---------|-----------------------------------------|---------------------|
| (Hp)                             |                       |                                                |                              |                         |         |                                         |                     |
| <i>hMni</i>                      | NICD_reg              | <i>NICD</i> activation K <sub>0.5</sub>        | <i>ps_ni</i> (default 2000)  | context dependent       | scanned | phenomenological                        | §1.1                |
| <i>Kdni</i>                      | NICD_reg              | <i>NICD</i> degradation rate                   | <i>ps_Kdni/4</i> (default 0) | context dependent       | scanned | experimentally constrained              | §1.1                |
| <i>tmDN</i>                      | CC3D → <i>NICD</i>    | Trans <i>DLL4</i> – <i>NOTCH1</i> interactions | computed each MCS            | 0–5000                  | dynamic | structural (coupling)                   | EC_Connect_core 235 |
| <i>cmDN</i>                      | cis_interactions      | Productive cis <i>DLL4</i> – <i>NOTCH1</i>     | computed                     | 0–5000                  | dynamic | structural (coupling)                   | §2.4                |
| <b>Notch / ligand expression</b> |                       |                                                |                              |                         |         |                                         |                     |
| <i>bN1</i>                       | ligands_receptors_reg | Basal <i>NOTCH1</i> production                 | 250                          | —                       | fixed   | phenomenological                        | §2.1                |
| <i>Kdn1</i>                      | ligands_receptors_reg | <i>NOTCH1</i> degradation                      | 0.05                         | —                       | fixed   | phenomenological/literature constrained | §2.1                |
| <i>hMn1</i>                      | ligands_receptors_reg | <i>NOTCH1</i> inducibility K <sub>0.5</sub>    | 50                           | —                       | fixed   | phenomenological                        | §2.1                |
| <i>N1</i>                        | ligands_receptors_reg | <i>NOTCH1</i> level                            | dynamic                      | 0–5000                  | dynamic | structural (state)                      | §2.1                |
| <i>bD4</i>                       | ligands_receptors_reg | Basal <i>DLL4</i> production                   | 250                          | —                       | fixed   | phenomenological                        | §2.2                |
| <i>hMd4</i>                      | ligands_receptors_reg | <i>DLL4</i> repression K <sub>0.5</sub>        | 50                           | —                       | fixed   | phenomenological                        | §2.2                |
| <i>Kdd4</i>                      | ligands_receptors_reg | <i>DLL4</i> degradation                        | 0.05                         | —                       | fixed   | phenomenological/literature constrained | §2.2                |
| <i>dD4</i>                       | CC3D → Antimony       | Engaged <i>DLL4</i> loss term                  | computed each MCS            | 0– <i>D4</i> -dependent | dynamic | structural (coupling)                   | EC_Connect_core 235 |
| <i>D4</i>                        | ligands_receptors_reg | <i>DLL4</i> level                              | dynamic                      | 0–5000                  | dynamic | structural (state)                      | §2.2                |
| <i>VmJ1</i>                      | ligands_receptors_reg | Max <i>Jagged1</i> induction                   | <i>amp * ps_J1ind</i>        | context dependent       | scanned | phenomenological                        | §2.3                |
| <i>hMj1</i>                      | ligands_receptors_reg | <i>Jagged1</i> induction K <sub>0.5</sub>      | <i>ps_ja</i>                 | context dependent       | scanned | phenomenological                        | §2.3                |
| <i>bJ1</i>                       | ligands_receptors_reg | Basal <i>Jagged1</i>                           | 0                            | —                       | fixed   | phenomenological                        | §2.3                |

| Parameter              | Module                | Species / Process            | Value in notebook                | Typical range  | Status  | Basis                                   | Section / Code     |
|------------------------|-----------------------|------------------------------|----------------------------------|----------------|---------|-----------------------------------------|--------------------|
| production             |                       |                              |                                  |                |         |                                         |                    |
| Kdj1                   | ligands_receptors_reg | Jagged1 degradation          | 0.05                             | —              | fixed   | phenomenological/literature constrained | §2.3               |
| dJ1                    | CC3D → Antimony       | Engaged Jagged1 loss term    | computed each MCS                | 0–J1-dependent | dynamic | structural (coupling)                   | EC_Connect_core235 |
| J1                     | ligands_receptors_reg | Jagged1 level                | dynamic                          | 0–5000         | dynamic | structural (state)                      | §2.3               |
| Trans interactions     |                       |                              |                                  |                |         |                                         |                    |
| tmJN                   | CC3D → NICD           | Trans Jagged1–NOTCH1         | computed each MCS                | 0–5000         | dynamic | structural (coupling)                   | EC_Connect_core235 |
| Rd4                    | CC3D                  | Reacted DLLA                 | computed                         | 0–5000         | dynamic | structural (coupling)                   | EC_Connect_core235 |
| Rj1                    | CC3D                  | Reacted Jagged1              | computed                         | 0–5000         | dynamic | structural (coupling)                   | EC_Connect_core235 |
| Cis interactions       |                       |                              |                                  |                |         |                                         |                    |
| cr                     | cis_interactions      | Cis interaction rate scaling | 0                                | 0–1            | scanned | phenomenological                        | §2.4               |
| crd                    | cis_interactions      | Effective cis rate (derived) | $cr * (1 - tmDN / (300 + tmDN))$ | 0–cr           | dynamic | structural (coupling)                   | §2.4               |
| kcd                    | cis_interactions      | Cis complex degradation      | 0.10                             | 0.05–0.2       | fixed   | phenomenological                        | §2.4               |
| kconv (ps_kconv)       | cis_interactions      | cDN → cmDN conversion rate   | 0                                | 0–0.5          | scanned | phenomenological                        | §2.4               |
| cDN                    | cis_interactions      | Cis DLL4–N1 complex          | dynamic                          | 0–5000         | dynamic | structural (state)                      | §2.4               |
| cmDN                   | cis_interactions      | Active cis DLL4–N1           | derived                          | 0–5000         | dynamic | structural (coupling)                   | §2.4               |
| cJN                    | cis_interactions      | Cis Jagged1–N1 complex       | dynamic                          | 0–5000         | dynamic | structural (state)                      | §2.4               |
| Competition parameters |                       |                              |                                  |                |         |                                         |                    |

| Parameter                              | Module                | Species / Process                             | Value in notebook | Typical range | Status     | Basis                      | Section / Code         |
|----------------------------------------|-----------------------|-----------------------------------------------|-------------------|---------------|------------|----------------------------|------------------------|
| kpDJ                                   | CC3D                  | <i>DLA4</i><br><i>Jagged1</i><br>partitioning | computed each MCS | 0–1           | dynamic    | phenomenological control   | EC_Connect_core<br>235 |
| KJC                                    | ligands_receptors_reg | <i>Jagged1</i><br>affinity<br>modifier        | 1.0               | 0.5–2         | fixed      | phenomenological control   | EC_Connect_core<br>235 |
| <b><i>HES1</i> – no autoregulation</b> |                       |                                               |                   |               |            |                            |                        |
| VmH1                                   | HES_no_auto           | Max <i>HES1</i><br>production                 | 10                | —             | fixed      | phenomenological           | §3.1                   |
| hMh1                                   | HES_no_auto           | <i>NICD</i> →<br><i>HES1</i> $K_{0.5}$        | 35                | —             | fixed      | phenomenological           | §3.1                   |
| Kdh1                                   | HES_no_auto           | <i>HES1</i><br>degradation                    | 0.09              | —             | fixed      | experimentally constrained | §3.1                   |
| bH1                                    | HES_no_auto           | Basal <i>HES1</i><br>production               | 0.0               | —             | fixed      | phenomenological           | §3.1                   |
| <b><i>HES1</i> – autoregulation</b>    |                       |                                               |                   |               |            |                            |                        |
| vm_prna                                | HES_auto              | Max pre-<br><i>mRNA</i><br>transcription      | 0.1               | 0.05–0.2      | scanned    | phenomenological           | §3.2.2                 |
| K                                      | HES_auto              | Shared $K_{0.5}$<br>(Hill/MM)                 | 0.5               | 0.3–0.7       | scanned    | phenomenological           | §3.2.2                 |
| hh1                                    | HES_auto              | <i>HES1</i><br>autorepression<br>Hill         | 8                 | 6–10          | fixed      | structural required        | §3.2.2                 |
| kmrp                                   | HES_auto              | <i>mRNA</i><br>processing<br>rate             | 0.1               | 0.05–0.2      | scanned    | structural required        | §3.2.2                 |
| kmrd                                   | HES_auto              | <i>mRNA</i><br>degradation                    | 0.05              | 0.02–0.1      | scanned    | structural required        | §3.2.2                 |
| kphp                                   | HES_auto              | Translation<br>rate                           | 0.1               | 0.05–0.2      | scanned    | structural required        | §3.2.2                 |
| khp                                    | HES_auto              | Protein<br>maturation                         | 0.1               | 0.05–0.2      | scanned    | structural required        | §3.2.2                 |
| khd                                    | HES_auto              | Active <i>HES1</i><br>degradation             | 0.075             | 0.05–0.15     | scanned    | experimentally constrained | §3.2.2                 |
| <i>NICD</i>                            | HES_auto              | <i>NICD</i> input<br>(scaled)                 | 0–1               | 0–1           | controlled | experimental driver        | §3.2.3                 |
| <b>Utility / kinetics</b>              |                       |                                               |                   |               |            |                            |                        |
| Hp()                                   | utility               | Positive Hill                                 | —                 | —             | structural | —                          | §0 (utility)           |
| Hc()                                   | utility               | Competitive<br>Hill                           | —                 | —             | structural | —                          | §0 (utility)           |

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|-----------|---------|-------------------|-------------------|---------------|------------|-------|----------------|
| $Hn()$    | utility | Negative Hill     | —                 | —             | structural | —     | §0 (utility)   |
| $Ma()$    | utility | Mass action       | —                 | —             | structural | —     | §0 (utility)   |
| $MM()$    | utility | Michaelis–Menten  | —                 | —             | structural | —     | §0 (utility)   |

Notes:

- In this model string, Parameter Scan knobs are explicitly declared ( $ps_*$ ).
- $NICD$  amplitude follows  $V_{mni}/K_{dni}$  when  $K_{dni}>0$ ; because  $K_{dni} := ps\_K_{dni}/4$ , amplitude/timescale depend on the chosen scan setting.
- $N1$ ,  $D4$ , and  $J1$  basals/decays place steady states in the  $10^3$ – $10^4$  regime (lumped units), consistent with Boareto et al PNAS 2015.
- Time unit in the Antimony layer is 1 min; characteristic times (e.g.,  $1/K_{dn1}$ ,  $1/K_{dd4}$ ,  $1/K_{dj1}$ ) are in real minutes.
- Typical ranges are provided where parameters are scanned or vary during simulations; fixed entries list “—”.